

Olist Seller Performance Analysis

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Identifying Key Success Factors of Sellers on Brazil's Largest E-Commerce Platform

Background & Problem Statement

The logo for Olist, featuring the word "olist" in a bold, blue, sans-serif font.

Olist is Brazil's largest e-commerce marketplace connecting thousands of sellers with millions of customers. Understanding seller performance is critical to sustaining platform growth and preventing top sellers from switching to competitors.

Analysis Questions:

1. Which sellers are most valuable based on revenue and order volume?
2. How do revenue and order trends change over time?
3. Which regions produce the best-performing sellers?
4. Which product categories dominate platform sales?

Methodology

Tools: SQL (SQLite) + Microsoft Excel (Power Query, Pivot Table, Pivot Chart)

Key Variables: seller_id, revenue, order volume, state, product category, payment_value, order_status, order_purchase_timestamp

Tables: sellers, orders, order_items, products, customers, order_payments

Data Source: Olist E-Commerce Public Dataset (Kaggle) · Period: Jan 2017 – Aug 2018

Steps:

1. Data Collection & Loading
2. Data Cleaning
3. Exploratory Data Analysis (EDA)
4. Data Visualization
5. Business Recommendations

Analysis Questions

Q1: Which sellers are most valuable by revenue & order volume?

Goal: Identify top contributors to platform revenue

Q2: How do monthly revenue & orders trend over time?

Goal: Understand platform growth patterns

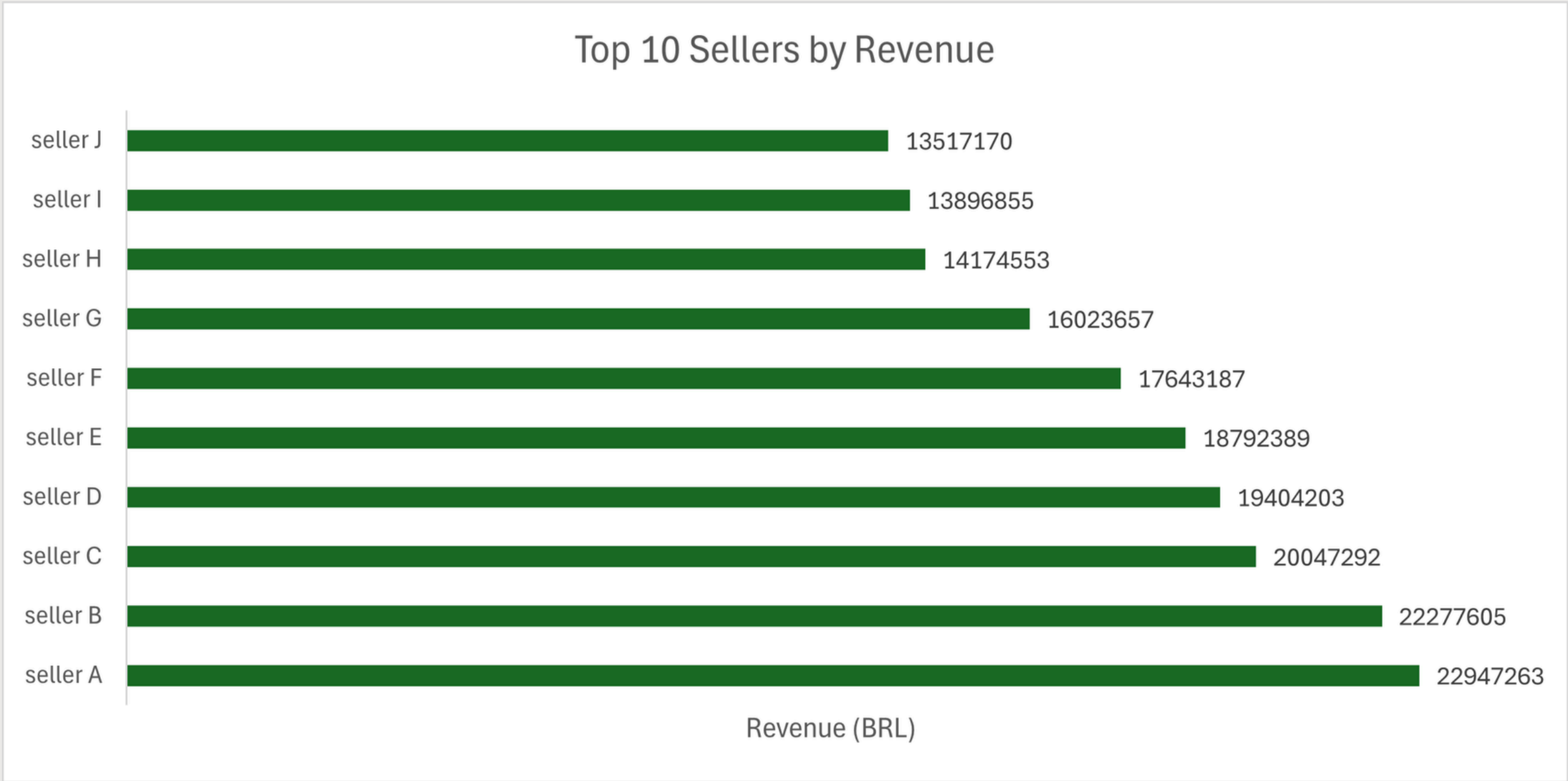
Q3: Which regions produce the best-performing sellers?

Goal: Determine priority areas for seller expansion and retention

Q4: Which product categories dominate platform sales?

Goal: Identify top categories for stock and promotion strateg

Top 10 Sellers by Revenue



Insight

Seller A leads with BRL 22.9M in revenue. The top 3 sellers contribute disproportionately more than ranks 4–10, revealing heavy revenue concentration among a small elite group.

Top 10 Sellers by Order Volume



Insight

Seller K dominates with 2,033 transactions yet doesn't appear in top revenue — indicating two distinct seller strategies: high-value low-volume vs. high-volume low-margin.

Monthly Revenue Trend



Insight

Revenue grew consistently from BRL 12M (Jan 2017) to a peak of BRL 101M (Nov 2017), then stabilized at BRL 85–99M throughout 2018 — reflecting Olist's rapid expansion phase followed by market maturity.

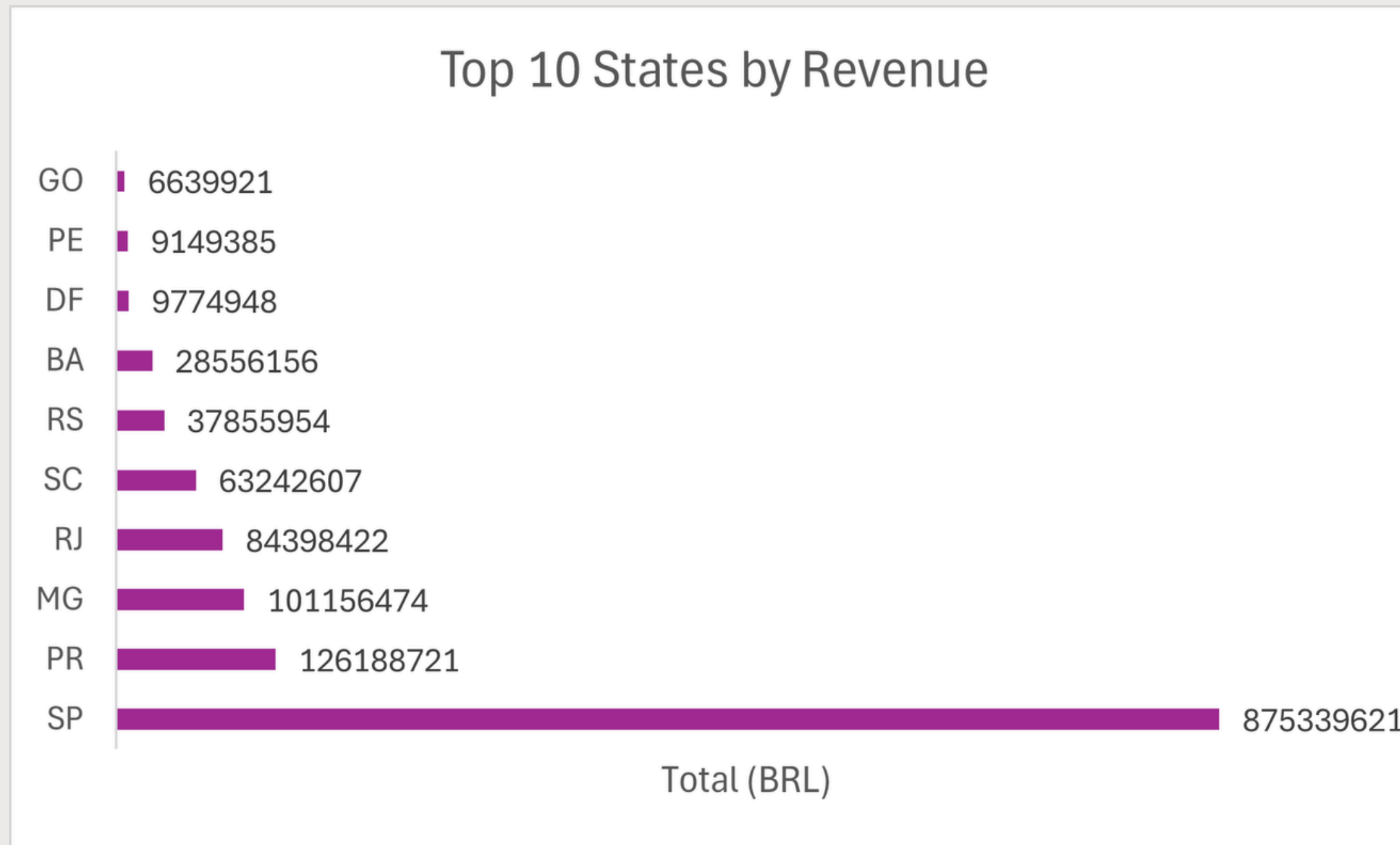
Monthly Order Volume Trend



Insight

Order volume peaked at 8,665 in Sep 2017 before stabilizing at 7,000–8,000 monthly. The parallel growth between volume and revenue confirms healthy platform scalability.

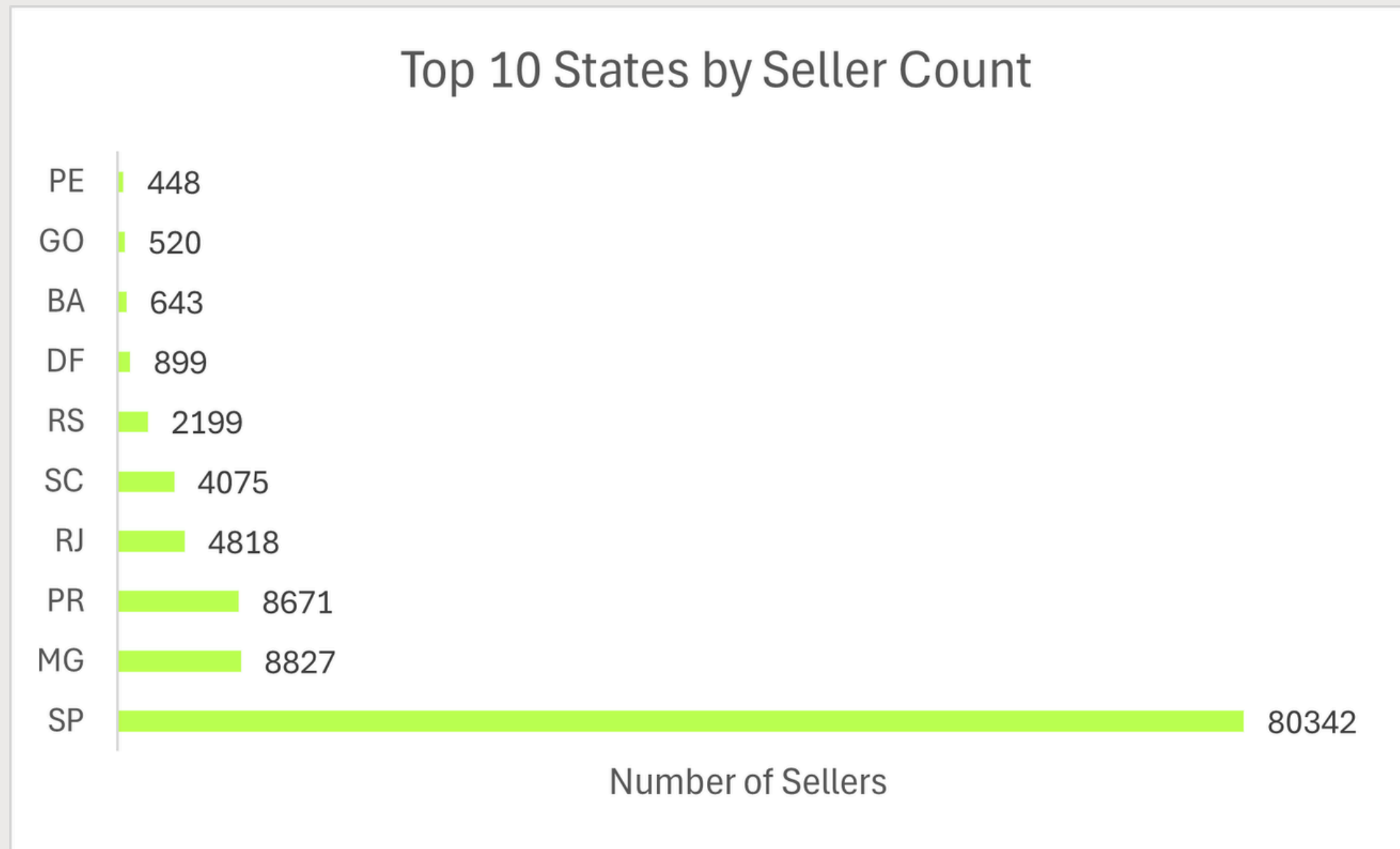
Seller Distribution by State (Revenue)



Insight

São Paulo dominates with BRL 875M — over 6x the second-ranked state (PR, BRL 126M), making SP the undisputed priority market for seller acquisition and retention.

Seller Distribution by State (Seller Count)

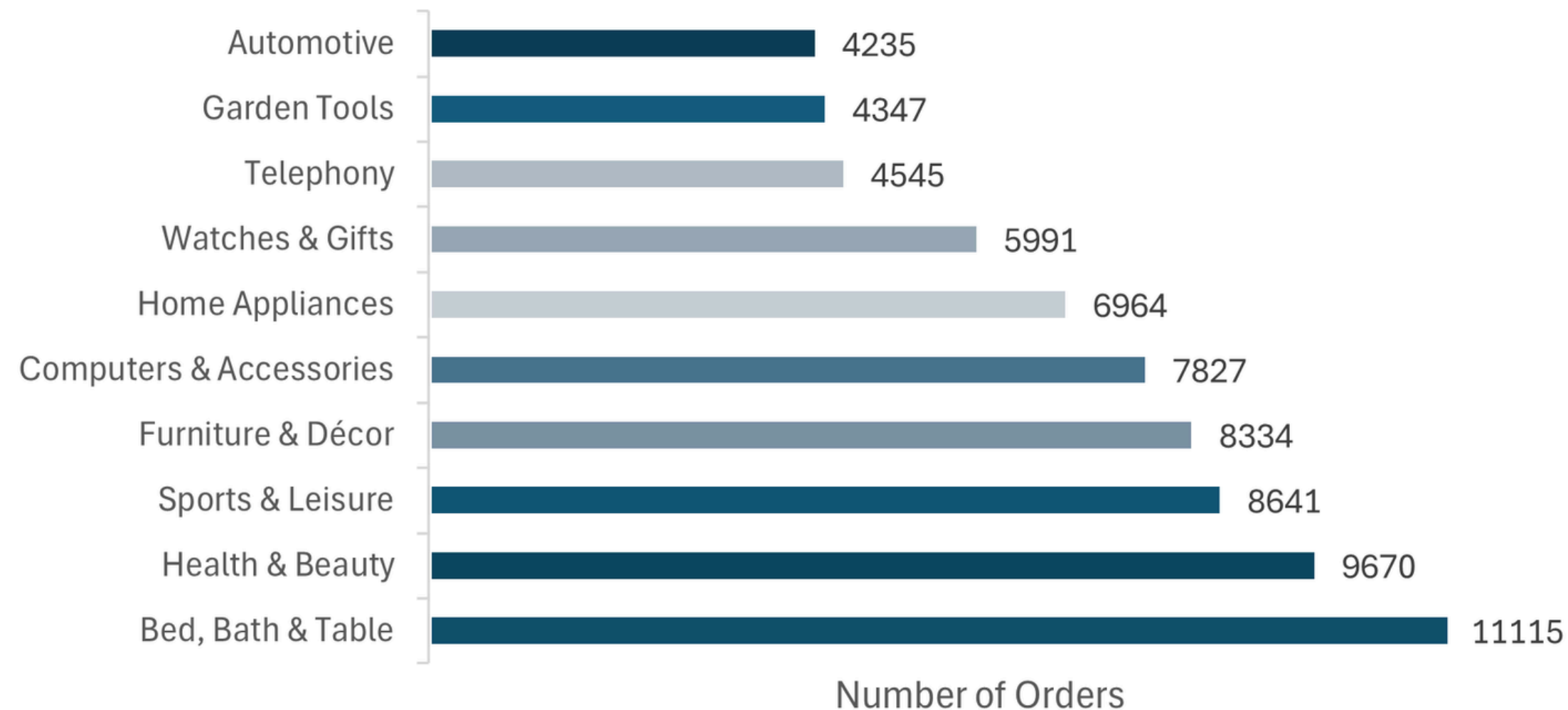


Insight

São Paulo leads with 80,342 sellers. However, MG and PR show strong seller counts relative to their revenue output, indicating untapped growth potential in both regions.

Product Categories by Order Volume

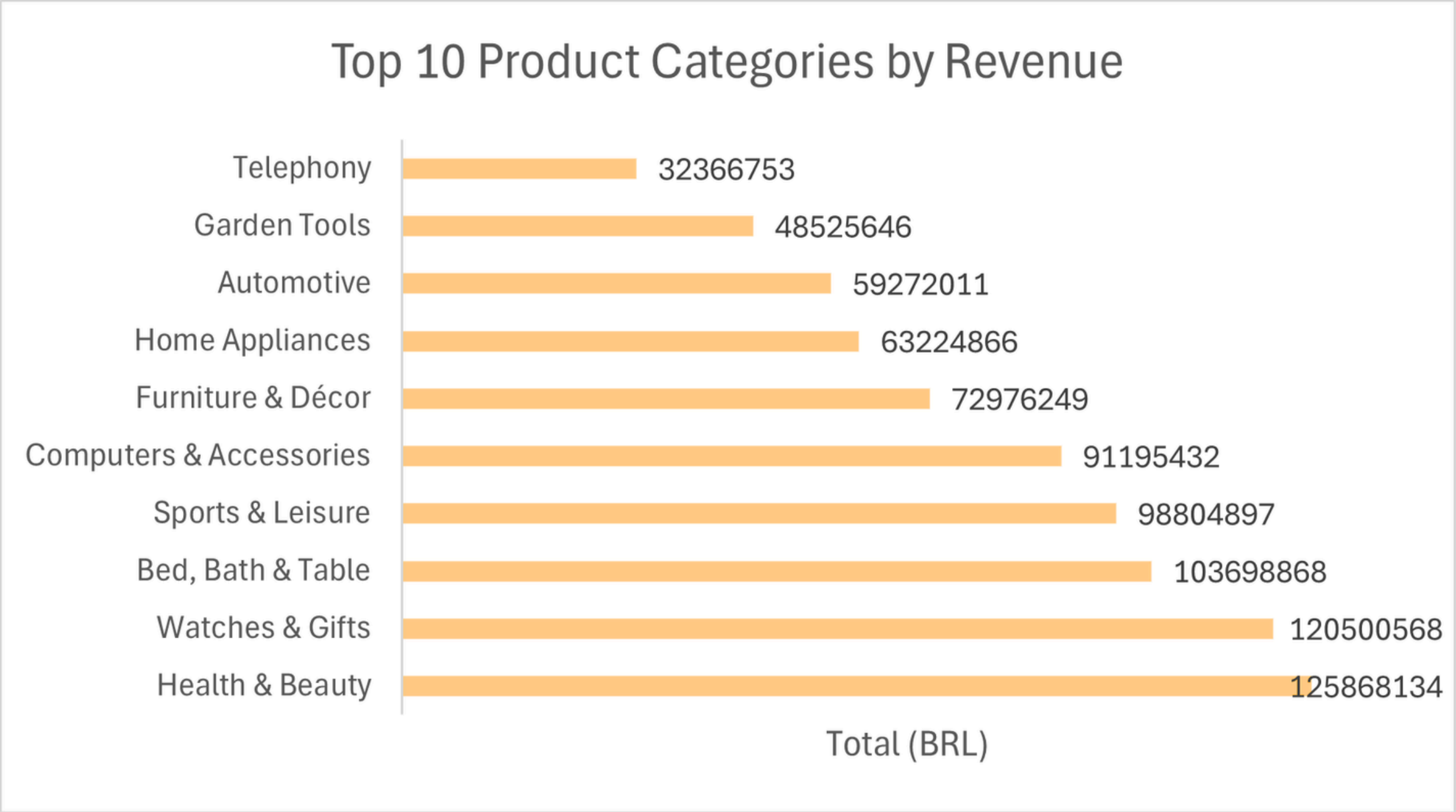
Top 10 Product Categories by Order Volume



Insight

Bed, Bath & Table leads with 11,115 orders, followed by Health & Beauty (9,670) and Sports & Leisure (8,641) — these three categories represent the highest consumer demand on the platform.

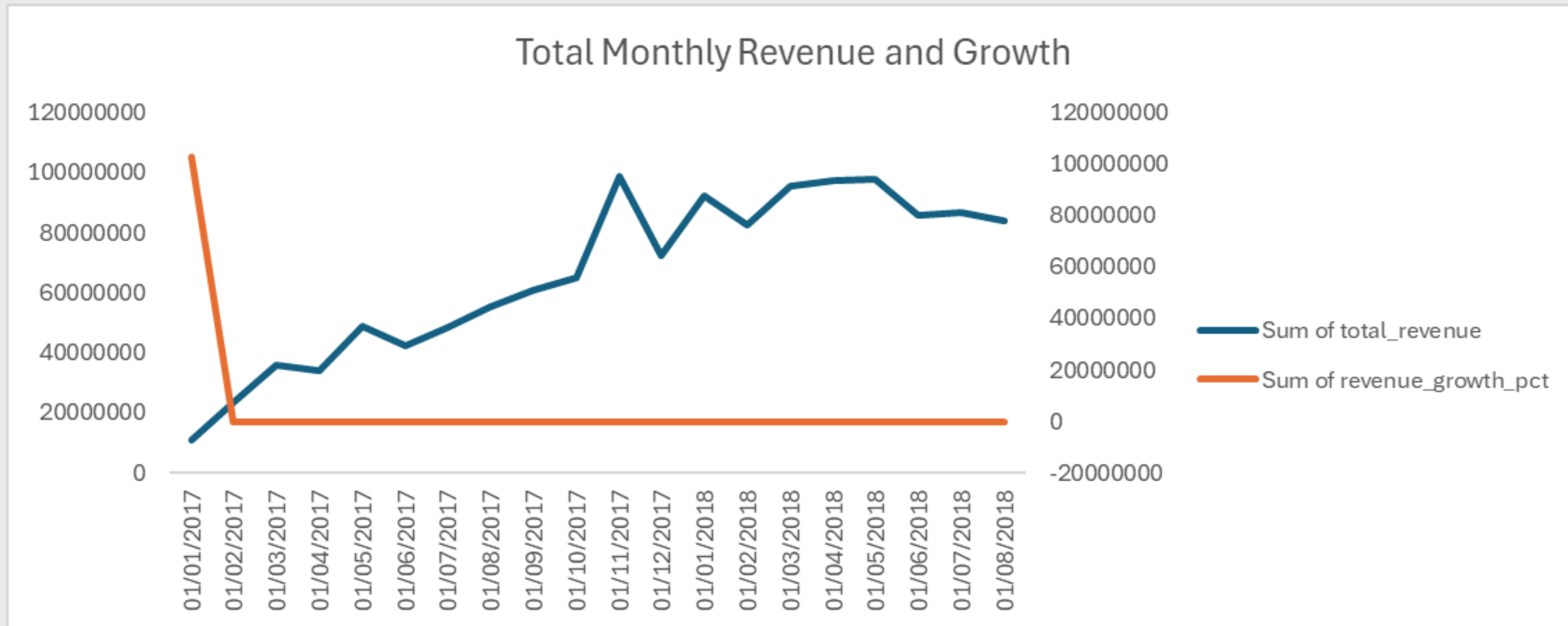
Product Categories by Revenue



Insight

Health & Beauty generates the highest revenue (BRL 125.8M) despite ranking second in order volume — signaling higher average transaction value and effective premium pricing strategy.

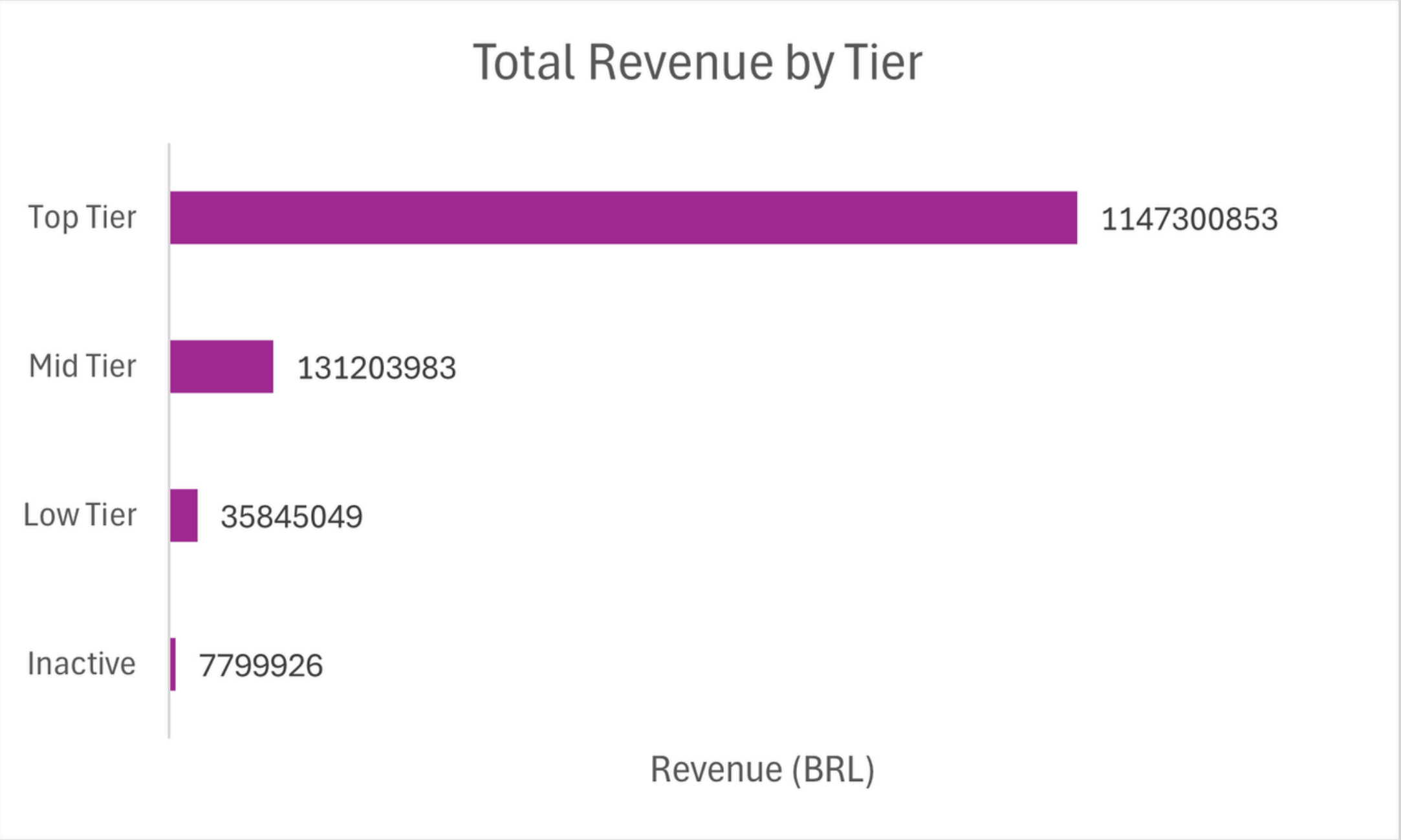
Total Revenue and Growth



Insight

Platform revenue grew consistently from BRL 12M (Jan 2017) to a peak of BRL 99M (Nov 2017), then stabilized at BRL 80–100M throughout 2018 — reflecting Olist's rapid expansion phase followed by market maturity

Total Revenue by Tier



Insight

Top Tier sellers account for 86% of total platform revenue (BRL 1.14B) despite representing only 25% of all sellers — a classic Pareto distribution pattern signaling high dependency risk on a small seller base.

Seller Cohort Retention Heatmap

Sum of activity_sellers		Column Labels																							
Row Labels		01/09/2016	01/10/2016	01/12/2016	01/01/2017	01/02/2017	01/03/2017	01/04/2017	01/05/2017	01/06/2017	01/07/2017	01/08/2017	01/09/2017	01/10/2017	01/11/2017	01/12/2017	01/01/2018	01/02/2018	01/03/2018	01/04/2018	01/05/2018	01/06/2018	01/07/2018	01/08/2018	Grand Total
01/09/2016		1	1																						2
01/10/2016			127		69	82	75	70	74	65	65	64	60	63	63	61	59	55	52	48	42	42	43	43	1322
01/12/2016				1	1																			2	
01/01/2017					149	106	105	90	99	86	77	87	77	80	78	64	71	66	69	66	56	56	50	57	1589
01/02/2017						214	135	120	124	100	111	111	98	95	91	98	83	82	81	87	69	71	65	57	1892
01/03/2017							161	97	91	85	79	80	75	69	80	62	65	56	55	52	50	51	43	46	1297
01/04/2017								111	61	49	57	53	51	42	50	43	43	35	39	35	33	28	30	26	786
01/05/2017									114	64	61	63	58	56	55	48	52	53	46	41	36	29	35	27	838
01/06/2017										70	30	28	33	23	27	27	24	18	21	21	17	15	17	16	387
01/07/2017											113	77	69	70	76	69	64	60	60	59	58	57	50	55	937
01/08/2017												120	66	57	69	55	57	62	55	50	49	49	40	42	771
01/09/2017													124	66	66	58	63	49	58	53	53	43	40	42	715
01/10/2017														141	97	79	83	67	71	69	67	56	61	48	839
01/11/2017															185	98	103	82	87	80	70	70	77	63	915
01/12/2017																84	51	46	46	41	37	33	31	31	400
01/01/2018																	138	80	70	80	63	65	56	51	603
01/02/2018																		117	63	61	53	50	57	50	451
01/03/2018																			108	67	59	58	49	48	389
01/04/2018																				195	121	105	104	94	619
01/05/2018																					172	98	99	95	464
01/06/2018																						189	107	114	410
01/07/2018																							184	104	288
01/08/2018																								152	152
Grand Total		1	128	1	219	402	476	488	563	519	593	683	711	762	937	846	956	928	981	1105	1105	1165	1238	1261	16068

Insight

Sellers who joined in Jan–Mar 2017 show the strongest long-term retention, maintaining 50–80 active sellers even 12+ months after their first transaction, suggesting early cohorts are the platform's most loyal contributors.

Business Recommendations

Top Seller Retention Program

Create loyalty incentives for top 10 sellers: lower commission fees or homepage priority placement

1

2

Regional Expansion

Focus seller acquisition in MG and PR which show strong seller count but underoptimized revenue

Category Optimization

Prioritize stock and promotions for Health & Beauty and Bed, Bath & Table

3

Q4 Seasonal Campaign

Revenue peaks in Nov 2017 — schedule major promotions every Q4

4

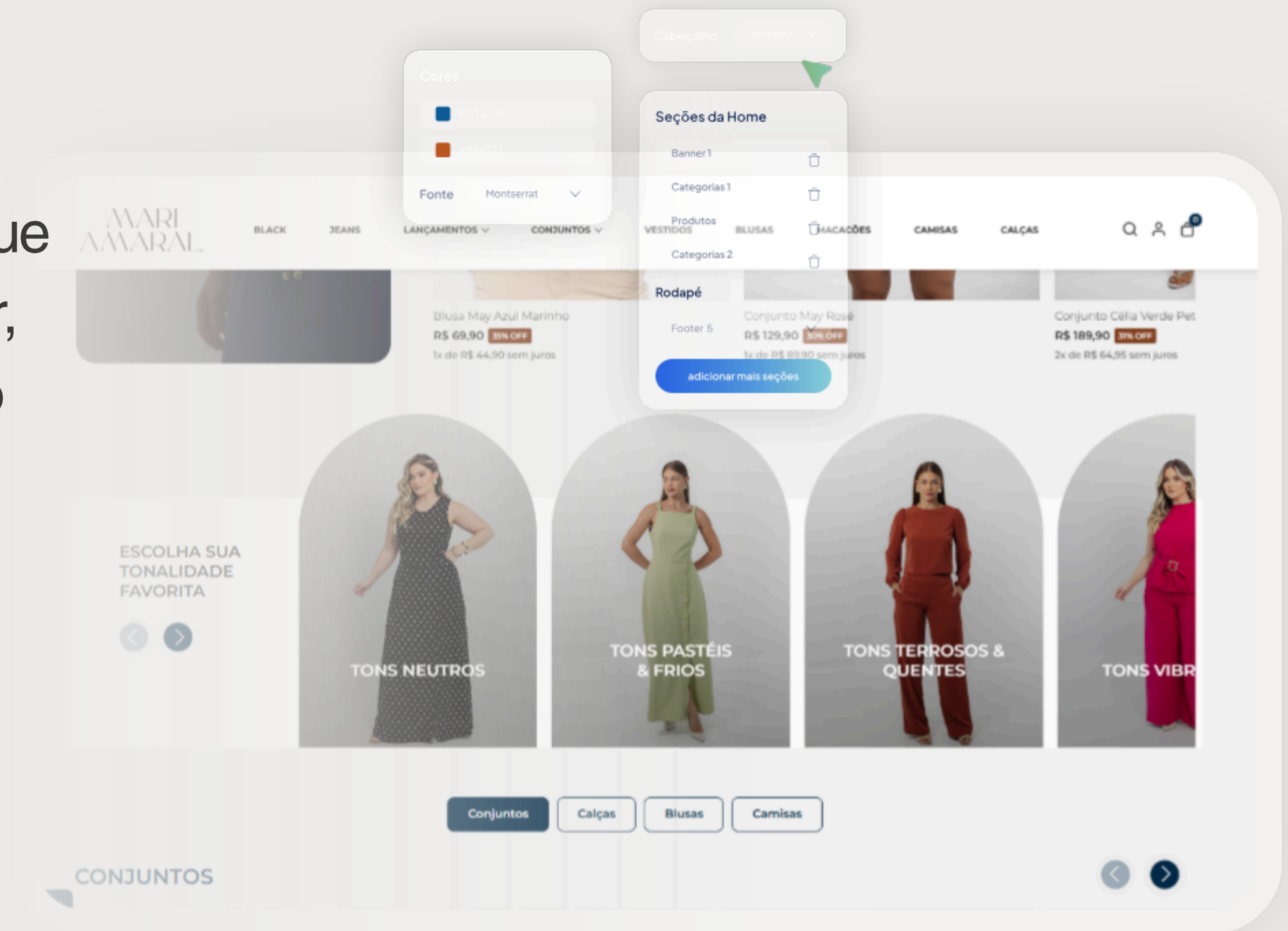
Seller Upselling Program

Push high-volume low-revenue sellers toward higher-value products through training

5

Conclusion

Based on the Olist dataset from January 2017 to August 2018, the platform demonstrated significant growth with revenue increasing nearly 8x in 20 months. However, performance is heavily concentrated in São Paulo and a small group of elite sellers. Health & Beauty and Bed, Bath & Table emerged as the strongest categories. Olist has a strong business foundation with substantial growth potential through expansion into underoptimized regions like Minas Gerais and Paraná.



The
End



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Thank You